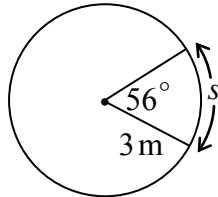


Question 1

2 marks

101

Use the information in the diagram to determine the value of the arc length “ s ”, given the central angle of 56° .



Question 2

4 marks

102

Solve $\tan^2 \theta - 5 \tan \theta + 4 = 0$ where $\theta \in \mathbb{R}$.

Question 3

4 marks

103

Solve:

$$2^{5x} = 3(5)^{x-3}$$

Question 4

3 marks

104

David and Sarah are in a class of 10 boys and 8 girls.

A committee of 3 boys and 2 girls is to be selected from the students in this class.

Determine the number of possible committees if David and Sarah cannot be on the same committee.

Question 5

3 marks

105

In the binomial expansion of $\left(\frac{3}{x^2} - x^5\right)^{10}$, simplify the 7th term.

Question 6

2 marks

106

A lake affected by acid rain has a pH of 4.4.

A person suffering from heartburn has a stomach acid pH of 1.2.

The pH of a solution is defined as $\text{pH} = -\log[H^+]$ where $[H^+]$ is the hydrogen ion concentration.

How many times greater is the hydrogen ion concentration of the stomach than that of the lake?

Express your answer as a whole number.

Note: A calculator is not required for the remaining test questions.

Question 7

4 marks

107

Solve the following equation algebraically over the interval $[0, 2\pi]$.

$$\cos 2\theta - 3\sin \theta - 2 = 0$$

Question 8

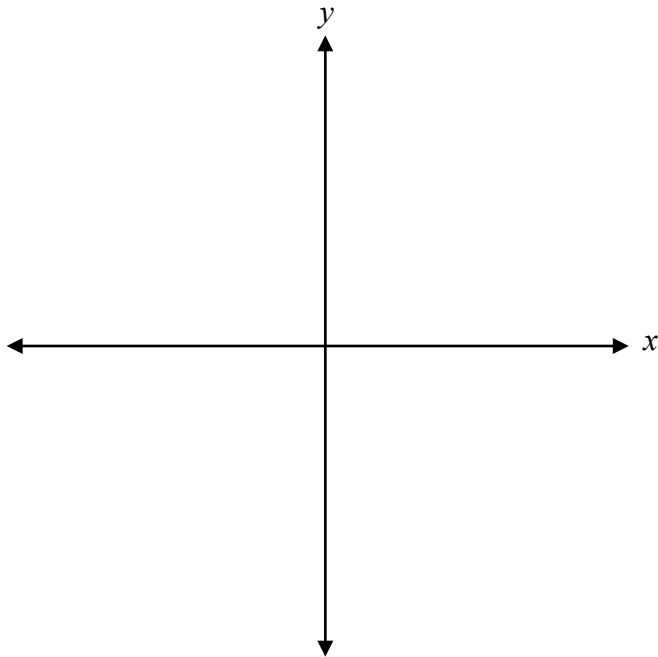
1 mark

108

Explain how the value of n affects the behaviour of the graph of the polynomial function

$p(x) = (x + 3)(x - 1)^n$, as $p(x)$ approaches the x -intercept at $x = 1$.

Sketch the angle -320° in standard position.

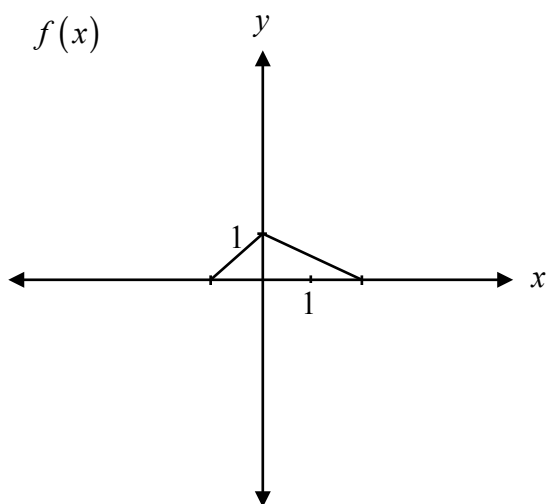


Question 10

1 mark

110

Given the graph of $y = f(x)$, explain how to graph $y = f(-x)$.



Question 11

1 mark

111

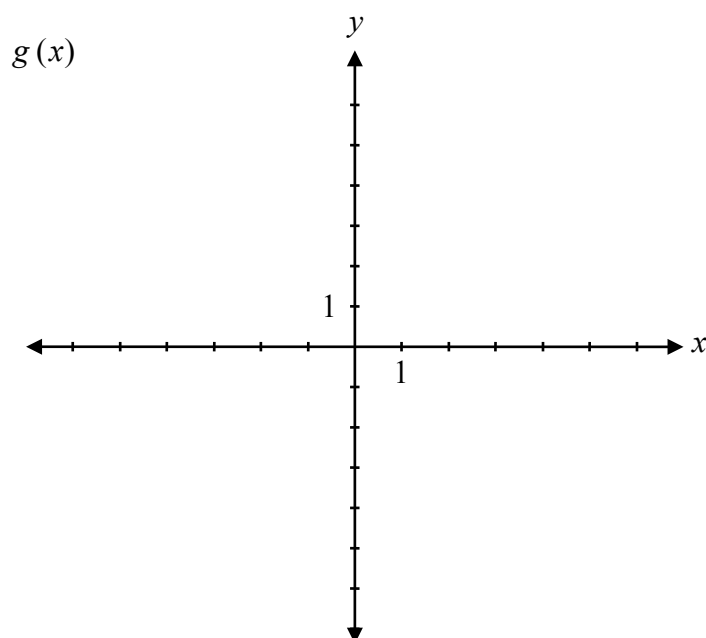
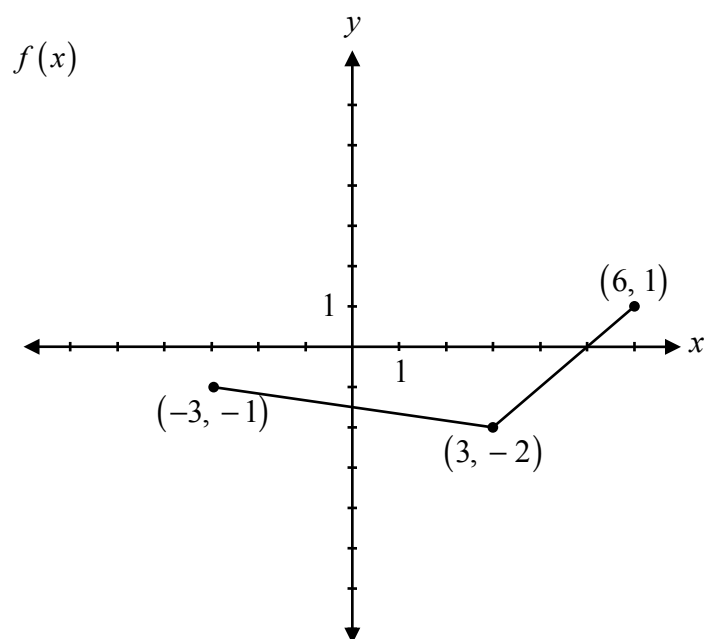
Explain how the graph of $y = \frac{3(x-1)}{(x-1)}$ is different than the graph of $y = 3$.

Question 12

2 marks

112

Given the graph of $f(x)$, sketch the graph of $g(x) = 2f(3x)$.

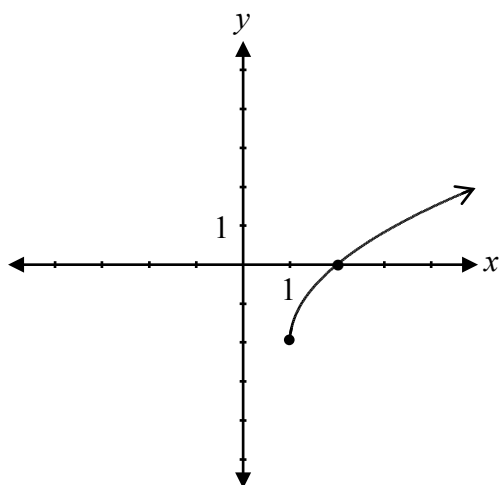


Question 13

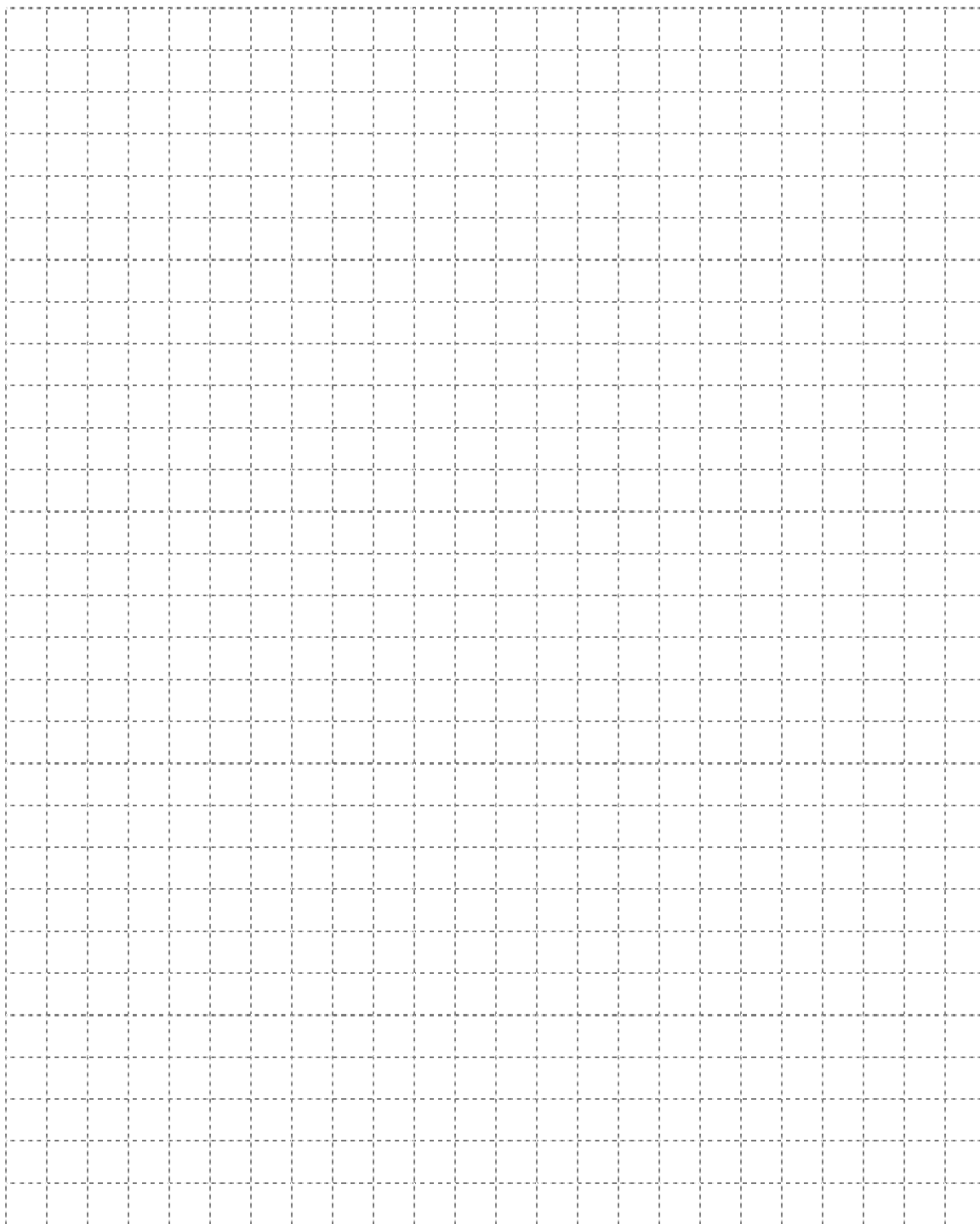
3 marks

113

Determine the equation of the radical function represented by the graph.



No marks will be awarded for work done on this page.



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